

Chapter 3: Motion in a Straight Line

3.1 Kinematic Equations

For a particle moving with uniform acceleration a , the velocity after time t is $v = u + at$ and the displacement is $s = ut + \frac{1}{2}at^2$. Eliminating t between these gives the third equation of motion, $v^2 = u^2 + 2as$. These equations are valid only when the acceleration is constant in both magnitude and direction.

KEY POINT

The area under a velocity–time graph gives the displacement of the particle over that interval.

3.2 Free Fall

Near the surface of the Earth, all bodies fall with the same acceleration $g \approx 9.8 \text{ m/s}^2$ in the absence of air resistance. A body dropped from rest covers a distance $h = \frac{1}{2}gt^2$ in time t .

Exercise 3A

Q1. A car starts from rest and accelerates uniformly at 2 m/s^2 for 5 s. The distance covered is

(A) 10 m (B) 25 m (C) 50 m (D) 5 m

Q2. A stone is dropped from a height of 45 m. Taking $g = 10 \text{ m/s}^2$, the time taken to reach the ground is

(A) 2 s (B) 3 s (C) 4.5 s (D) 9 s

Answer Key

Q1. (B) Q2. (B)

Corrigendum

The following corrections apply to this module but sit inside equations, figures or tightly set text that cannot be re-typeset in place. Markers [C#] in the margin point to each location where it could be found.

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FOR

Figure 3.4 arrow label

READ

should point downward

